Yuchen Xiao

Curriculum Vitae

Fakultät für Mathematik, Technische Universität Chemnitz
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Research Interests

Applied and Computational Harmonic Analysis
- Wavelets/framelets theory, discrete wavelet transform, fast Fourier transform, scientific computing.

Optimization and Approximation Theory
- Numerical analysis, image/signal processing, deep/machine learning.

Spherical Designs
- Construction and verification, application.

Education

Sep 2020 – Oct 2024 **PhD in Applied Mathematics**
City University of Hong Kong, Hong Kong, SAR China

Sep 2016 – Jun 2019 **MPhil in Computational Mathematics**
Jinan University, Guangzhou, China

Sep 2012 – Jun 2016 **BSc in Mathematics and Applied Mathematics**
Jinan University, Guangzhou, China

Employment

Sep 2024 – Sep 2026 **Postdoctoral Researcher**
Technische Universität Chemnitz, Chemnitz, Germany

Oct 2019 – Aug 2020 **Research Assistant**
City University of Hong Kong, Hong Kong, SAR China

Honours & Scholarships

Scholarships

2021–2022; 2023–2024 Research Tuition Scholarship, City University of Hong Kong

2016–2019 Scholarship for Postgraduate Students, The First Prize, Jinan University

2016 Entrance Scholarship for Excellent Recommendation Free Test Master's Degree, Jinan University

2016 Cambridge Academic Development Seminar Special Scholarship, Jinan University

2015–2016 Scholarship for Undergraduate Students, The Second Prize, Jinan University

2015 Undergraduate Innovative Talent Special Scholarship, Jinan University

Honours

2025 Outstanding Research Thesis Award (eq. *Summa cum laude*), City University of Hong Kong

2024 CSCI Excellent Paper Award, City University of Hong Kong

2016 The Best Graduation Thesis, Jinan University

2016 Outstanding Graduate Technology Innovation Award, Jinan University

Activities

Presentations

Aug 2023 10th International Congress on Industrial and Applied Mathematics (ICIAM), Tokyo

May 2023 International Conference on Applied Mathematics 2023 (ICAM), Hong Kong (invited)

Jan 2020 2020 Workshop on Optimal Configuration and Related Topics, Chengdu (invited)

Visits

Aug 2016 – Sep 2016 **Visiting Student**
University of Cambridge, Cambridge, Britain
Cambridge Academic Development Seminar Program

Publications

Journal Publications

2025 **Yuchen Xiao** and Xiaosheng Zhuang. Spherical designs for function approximation and beyond. *Sampling Theory, Signal Processing, and Data Analysis*, 23(2):9, 2025.

2023 **Yuchen Xiao** and Xiaosheng Zhuang. Spherical framelets from spherical designs. *SIAM Journal on Imaging Sciences*, 16(4):2072–2104, 2023.

2021 **Yuchen Xiao** and Xiaosheng Zhuang. Adaptive directional Haar tight framelets on bounded domains for digraph signal representations. *Journal of Fourier Analysis and Applications*, 27:1–26, 2021.

2020 Congpei An and **Yuchen Xiao**. Numerical construction of spherical t-designs by Barzilai-Borwein method. *Applied Numerical Mathematics*, 150:295–302, 2020.

Preprints

2026 Sebastian Neumayer, Daniel Potts, **Yuchen Xiao**, and Xiaosheng Zhuang. Adaptive Spherical Framelet-TV Regularization on Nested Spherical Designs. submitted

Conference Proceedings

2023 Xiaosheng Zhuang and **Yuchen Xiao**. Denoising on sphere via large spherical t-designs and spherical framelets. In *2023 International Conference on Sampling Theory and Applications (SampTA)*, pages 1–5, 2023.

2015 Jinlan Chen, **Yuchen Xiao**, and Jialiang Zhou. Comprehensive evaluation and prediction model of sustainable development. In *2015 Joint International Mechanical, Electronic and Information Technology Conference (JIMET-15)*, pages 565–569, 2015.

Projects & Competitions

Projects

2015–2016 The Dynamic Model of Adjusting Bus Arriving Time, Jinan University (PI)

2015–2016 Recommendation Algorithm of Mixed Commodity Information, Jinan University (Co-I)

2014–2015 The Dynamic Model of Predicting Bus Arriving Time, Jinan University (PI)

2014–2015 Commodity Information Pushing in Complex Networks, Jinan University (Co-I)

Competitions

2016 Postgraduate Academic Forum on Big Data and Supercomputing Technology, The Second Prize

2015 Jinan University Undergraduate Innovation Competition, The Entrance Prize

2015 Mathematical Contest In Modelling (MCM/ICM), The Honorable Mention

2014 China Undergraduate Mathematical Contest in Model (CUMCM), The Third Prize

2013 The Chinese Mathematics Competitions (CMC), The Second Prize

Grants

Internal

2023 UGC Conference Grants for ICIAM 2023

Academic Service

Referee

2026 Springer Nature, Foundations of Computational Mathematics

2024 American Mathematical Society (AMS), Mathematical Reviews (MR)

2024 International Joint Conference on Artificial Intelligence (IJCAI)

Membership

Since Jul 2023 Society for Industrial and Applied Mathematics (SIAM)

Teaching Experience

Sep 2024 – Sep 2026 **Technische Universität Chemnitz**

- A Mathematical Introduction to Learning Theory / Mathematische Grundlagen der Lerntheorie (26 SS)
- Mathematics for Science and Engineering II (26 SS)
- Mathematics for Science and Engineering I (25/26 WS)
- Inverse Problem / Inverse Probleme (25 SS)
- Mathematical Methods in Image Processing / Mathematische Methoden der Bildverarbeitung (24/25 WS)

Sep 2020 – Jun 2024 **City University of Hong Kong**

- Math Help Center (2020–2024)
- MA1200 - Calculus and Basic Linear Algebra I (20-21 Sem A-B, 22-23 Sem B)
- MA1201 - Calculus and Basic Linear Algebra II (20-21 Sem A, 21-22 Sem B)
- MA1300 - Enhanced Calculus and Linear Algebra I (22-23 Sem A)
- MA1501, GE1358 - Coordinate Geometry (23-24 Sem B)
- MA2172 - Applied Statistics for Sciences and Engineering (20-21 Sem A-B, 23-24 Sem A)
- MA2185 - Discrete Mathematics (21-22 Sem A)
- MA2507 - Computing Mathematics Laboratory (22-23 Sem B)
- MA2506, MA2510 - Probability and Statistics (20-21 Sem B)

Sep 2016 – Jul 2018 **Jinan University**

- Linear Algebra
- Calculus
- Topology and Abstract Algebra

Skills

Software

LaTeX, MATLAB, Linux (Shell), R, Python, C, Maple and Adobe Photoshop.

Languages

Mandarin, Cantonese, English and German (beginner).

Interests

Photography, cooking, guitar, soccer and taekwondo.